PROJECT WALKTHROUGH

Online Retail Analysis
From source data to customer insights

Prepared 12 November 2026 | Based on saved project results

What we achieved: a working, reproducible exploratory data analysis (EDA) project that downloads real retail transactions, cleans them, calculates business metrics, segments customers, and produces tables, charts, and an executed notebook.

Final result	Value
Source transaction lines	541,909
Valid positive sales lines	524,878
Positive sales revenue before refunds	GBP 10,642,110.80
Distinct sales invoices	19,960
Known purchasing customers	4,338
Deliverables	11 charts, 6 CSV tables, summary JSON, executed notebook

Where we started: the code and notebook were largely written, but the dataset and charts were missing, dependencies were absent from the virtual environment, and there were no saved notebook results or automated tests.

Where we are now: the core EDA runs on the real workbook. Reliability issues were corrected, 10 regression tests pass, all 11 notebook code cells have executed, and analytical totals reconcile across outputs.

EDA means exploring and explaining historical data. This project produces descriptive evidence; it does not train a forecasting or churn model.

1. The technology stack

Each tool has a specific job in the workflow. Python runs the analysis; the notebook explains it; CSV, JSON, and PNG files make the results reusable.

Technology	What we used it for
Python 3.12.14	Validated project runtime and reusable analysis functions.
Pandas	Excel loading, cleaning, grouped sales metrics, RFM and cohort tables.
NumPy	Numerical checks, score bands, conditional segment labels and missing values.
Matplotlib + Seaborn	11 saved charts: trends, rankings, distributions, correlations and retention.
OpenPyXL	Reading the original .xlsx workbook through Pandas.
Jupyter + IPython kernel	A readable analysis notebook with saved tables, metrics and chart outputs.
nbformat + nbclient	Notebook structure validation and automated execution of every code cell.
unittest + unittest.mock	10 regression tests, including controlled download failure scenarios.
pathlib, urllib, zipfile, json	Portable paths, source downloads, archive handling and metrics export.
VS Code, .venv, uv and pip	Editing, isolated dependency setup, installation and compatibility checks.
requirements-lock.txt	Exact tested package versions for repeatable installation.

Report tools: ReportLab generates this PDF; pypdf checks its pages and extractable text. They are optional reporting utilities, separate from the EDA's requirements files.

No database, cloud deployment, dashboard or machine-learning model was needed for the completed core analysis.

2. How the data became usable

Source: the UCI Online Retail Excel workbook, covering 1 December 2010 to 9 December 2011. Its eight fields describe invoice, stock code, description, quantity, invoice time, unit price, customer ID and country.

Cleaning stage	Rows	Meaning
Original workbook	541,909	Source remains unchanged.
Exact duplicates removed	5,268	Assumed repeated records; identical legitimate lines cannot be distinguished.
Deduplicated records	536,641	Basis for the sales filter and return summary.
Invalid or cancelled lines excluded	11,763	Combined exclusion count; overlapping categories are counted once.
Valid sales view	524,878	Positive quantity and price, valid date and identifiers, finite line value.

Revenue calculation: TotalPrice = Quantity x UnitPrice. Sales revenue sums valid positive line values. Invoice counts use distinct invoice numbers; average invoice value is total sales divided by the invoice count.

Anonymous purchases: missing customer IDs are retained in sales totals. The 132,186 valid anonymous lines are excluded only from customer-level analysis; IDs are never invented or filled in.

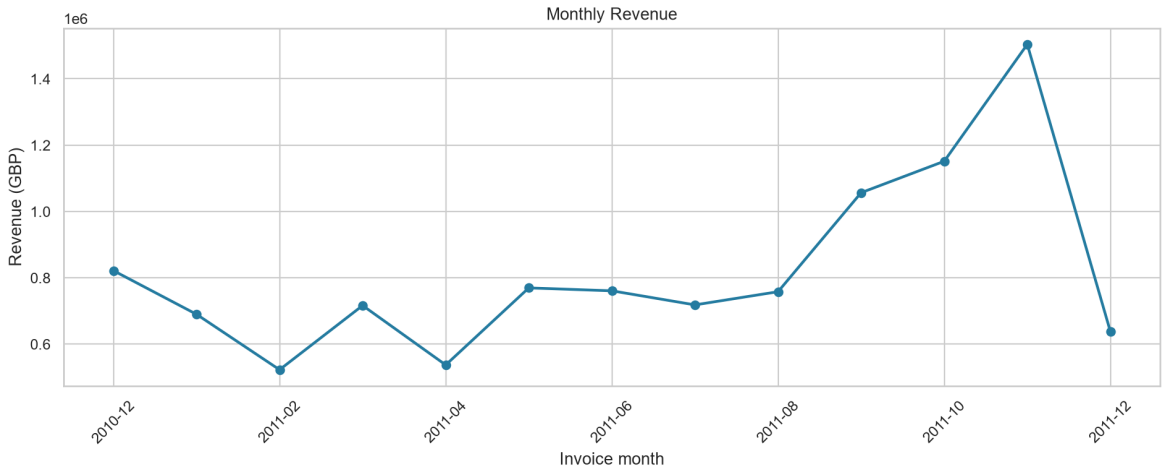
Returns: credit-note invoices and negative-quantity rows are summarized separately after deduplication. Their categories overlap and must not be added together. This project has not matched individual refunds to original orders.

Reliability fixes: downloads are written to temporary files before being published, corrupt archives trigger recovery, workbook selection is explicit, required columns are checked, and only the expected workbook is extracted.

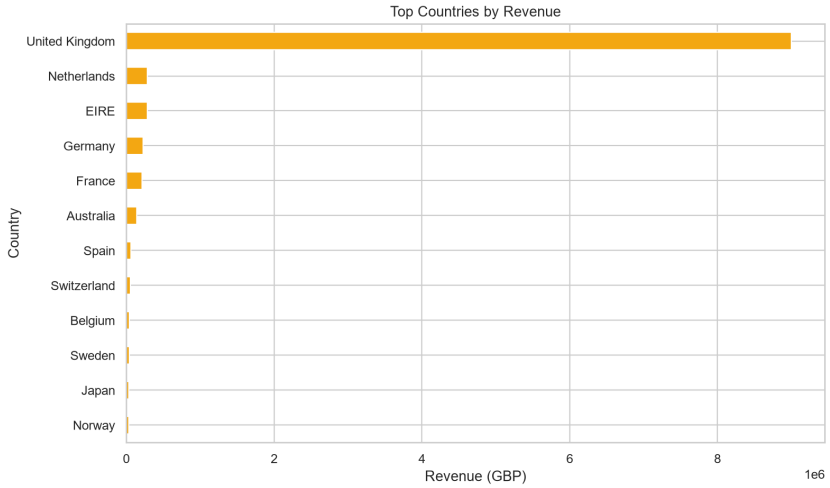
Dataset citation: Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. doi:10.24432/C5BW33. CC BY 4.0. Source attribution is also in README.md.

3. Sales results and what they mean

Average invoice value is **GBP 533.17**. The United Kingdom contributes **84.59%** of positive sales. November 2011 leads with **GBP 1,503,866.78**.



December 2011 contains only the first nine days. The final drop is not evidence of a full-month sales decline. One observation window is insufficient to establish recurring seasonality.



Revenue concentration: 803 of 3,922 stock codes (20.47%) account for at least 80% of positive sales revenue. Stock codes, rather than descriptions, define products; their most common description supplies the label.

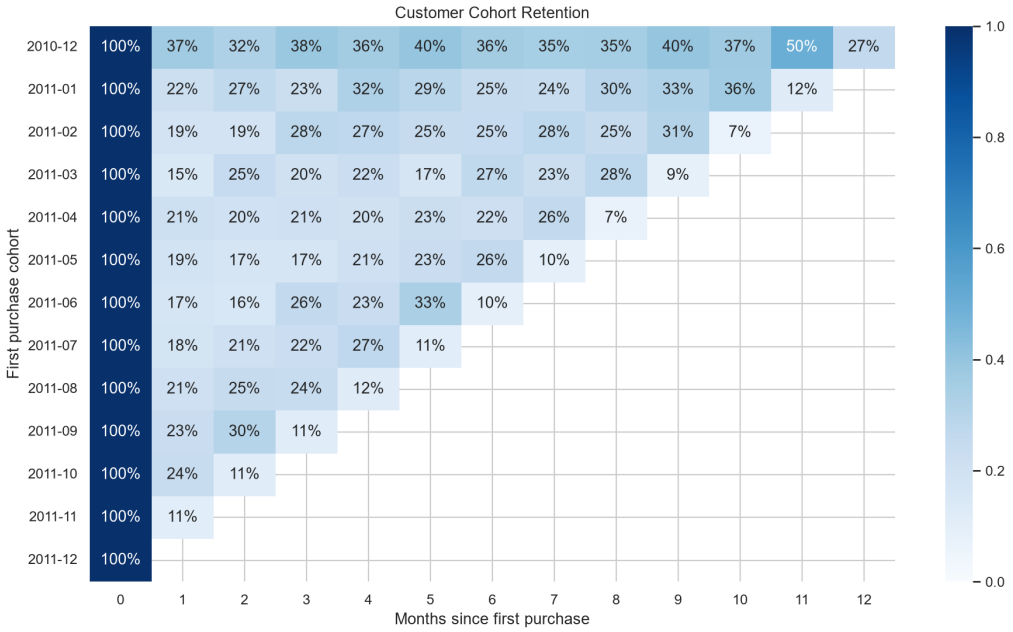
Practical interpretation: investigate UK operations and leading codes first. The highest-revenue code is DOTCOM POSTAGE (GBP 206,248.77), a service charge. Separate service items and investigate exceptional orders and refunds before making merchandise or stock recommendations.

4. Customer behavior and retention

65.58% of known purchasing customers placed more than one valid invoice during the observed window. This is a historical repeat purchase share, not a forecast of future loyalty.

RFM explained: Recency asks how recently someone bought, Frequency counts distinct invoices, and Monetary adds their positive sales. Four score bands use average percentile ranks so tied values receive identical scores. The reference date is the day after the last valid purchase in all sales.

Exploratory segment	Customers
Needs attention	1,959
Champions	1,299
At risk	565
Loyal customers	515



Each row groups customers by their first observed purchase month. A cell shows the share who purchased in that later month. Month zero is 100%; observed months with no returning customers are zero; future months stay blank. The final activity month is partial, and first observed purchase may not be true acquisition.

Next action: use these segments to design retention experiments and measure actual outcomes. Segment labels are simple exploratory rules, not validated churn predictions or campaign performance estimates.

5. Deliverables, validation and next steps

File or folder	What you can use it for
src/data_cleaning.py	Run the complete reusable pipeline.
notebooks/online_retail_analysis.ipynb	Follow the executed walkthrough and see saved findings.
dataset/	Original workbook and six output CSVs: cleaned sales, RFM, returns, products, monthly sales and cohorts.
visualizations/	11 PNG charts for trends, products, countries, customers and retention.
analysis_summary.json	Computed headline metrics and cleaning audit.
tests/test_analysis.py	Regression checks for key analytical rules and download recovery.
README.md + requirements files	Methodology, findings, setup instructions and tested dependencies.

Validation completed: all 10 regression tests pass; all 11 notebook code cells executed without errors; installed dependencies passed compatibility checks. Sales totals reconcile across cleaned transactions, monthly and product outputs. RFM revenue matches known-customer sales; customer and segment counts agree; cohort values stay between zero and one. All 11 charts were visually reviewed.

Run it again in VS Code: select .venv/Scripts/python.exe as interpreter and notebook kernel. From the project root, run the following commands or open the notebook and choose Run All.

```
python src/data_cleaning.py
python -m unittest discover -s tests -v
python -m pip check
```

Development priorities: (1) classify merchandise versus service codes and reconcile credit notes to sales; (2) add configurable paths and a dashboard if needed; (3) obtain more periods before exploring forecasting, seasonality or customer lifetime value.

Final outcome: a completed and locally validated core EDA, with reusable outputs and documented assumptions. Positive sales are before refunds, not net revenue or profit. Product costs and campaign data are absent, so profitability and marketing effectiveness remain outside the evidence available.